

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

Complete

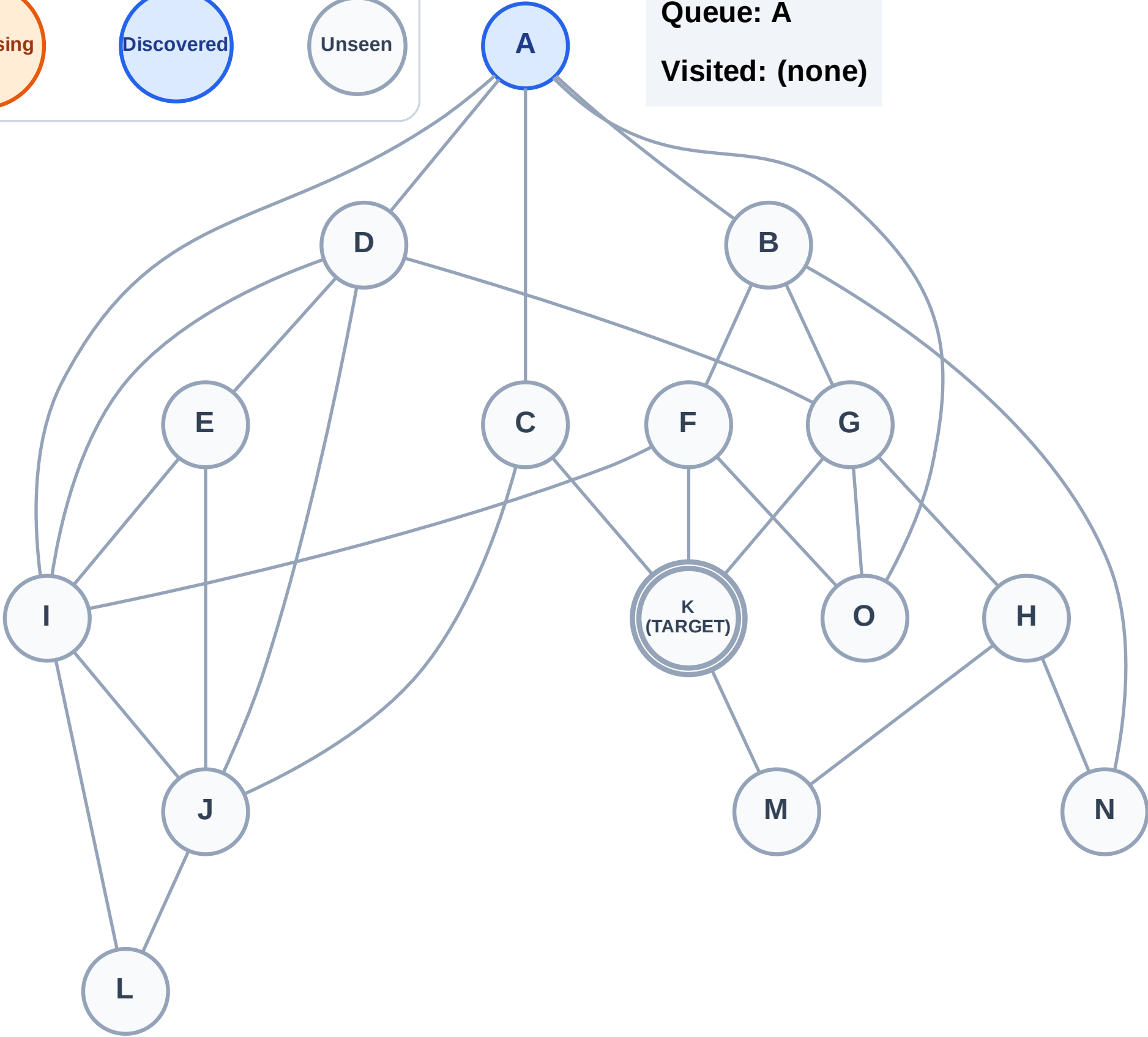
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

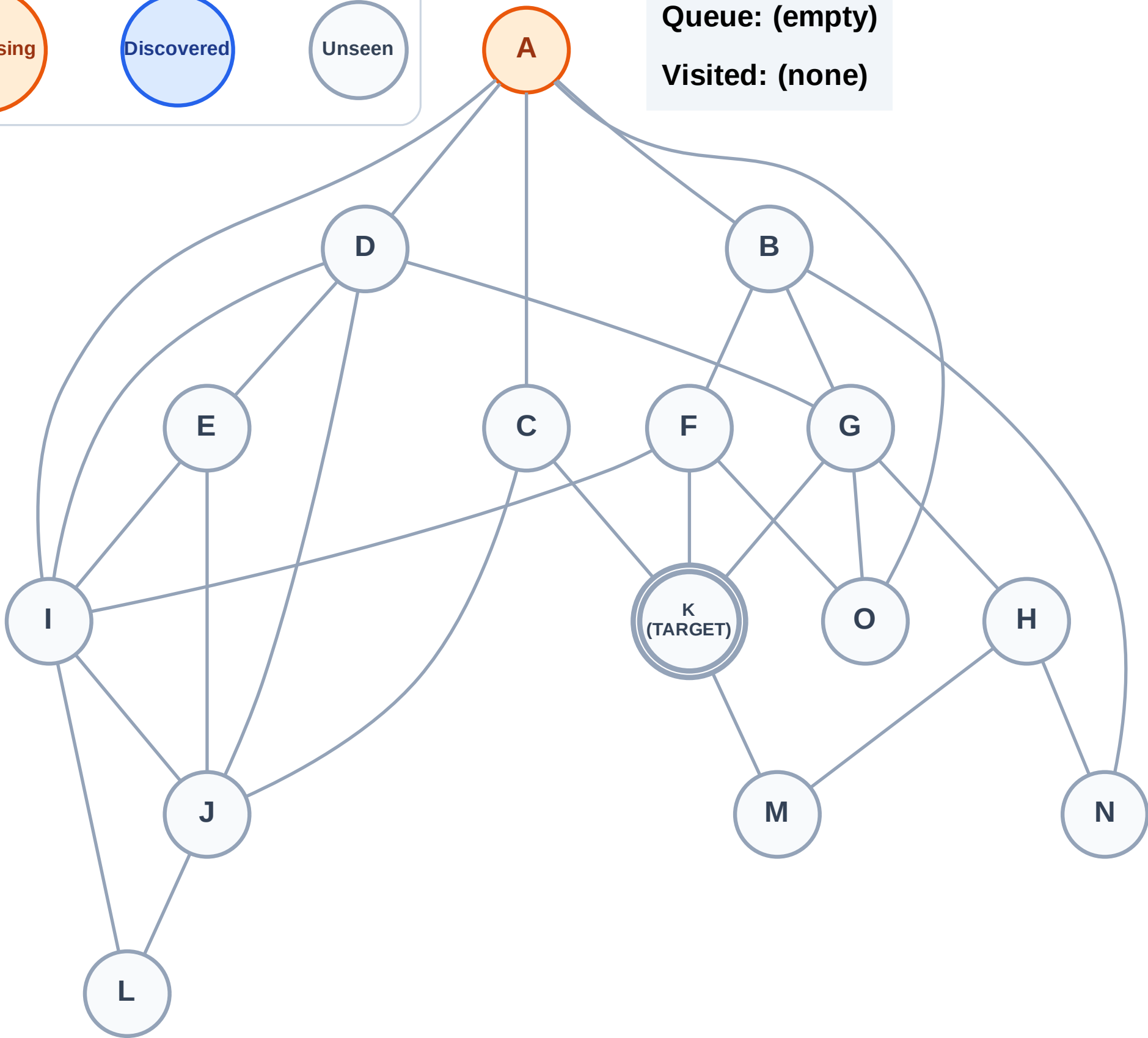
Complete

Processing

Discovered

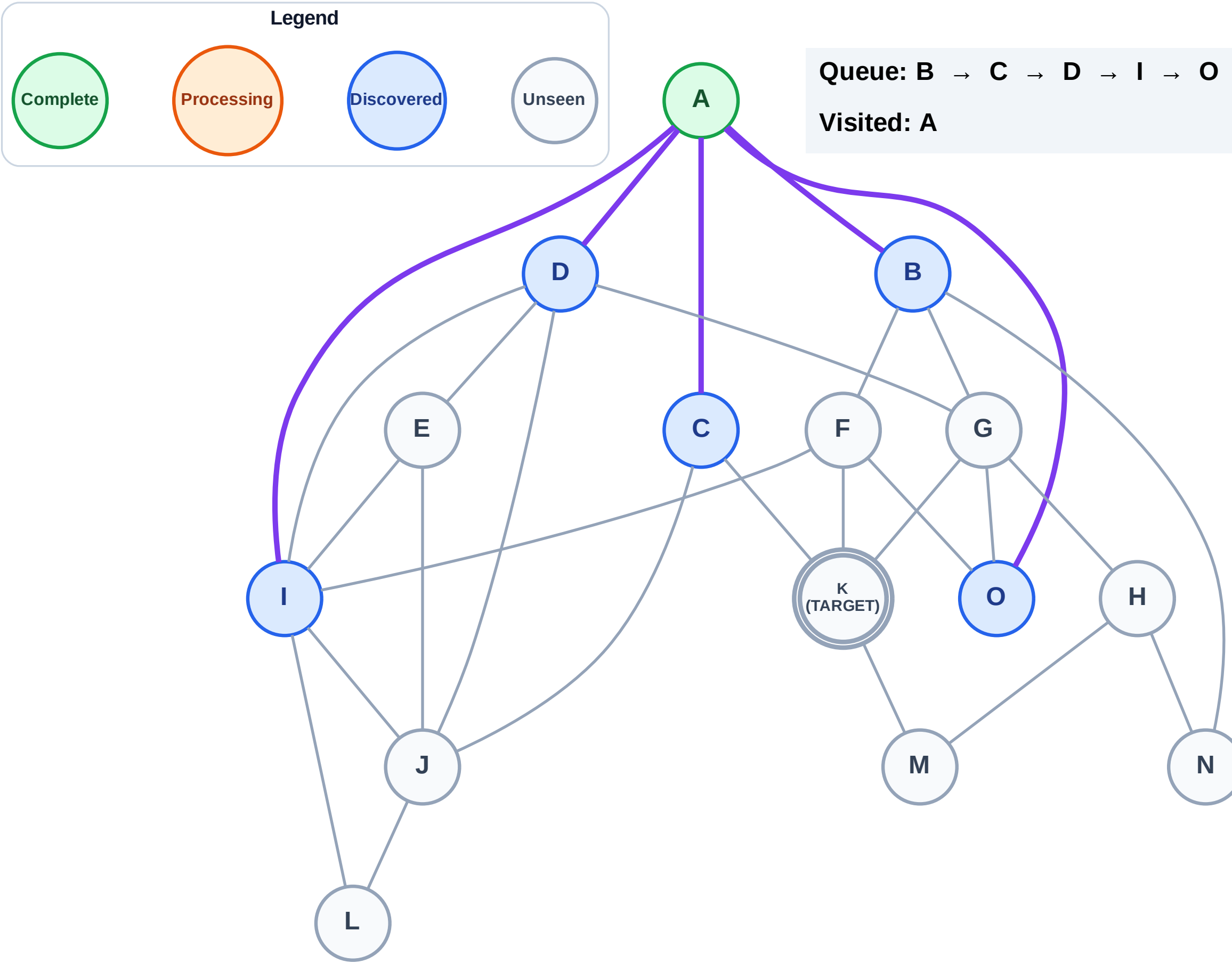
Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover B, C, D, I, O from A



BFS Traversal

Step 4: Process node B

Legend

Complete

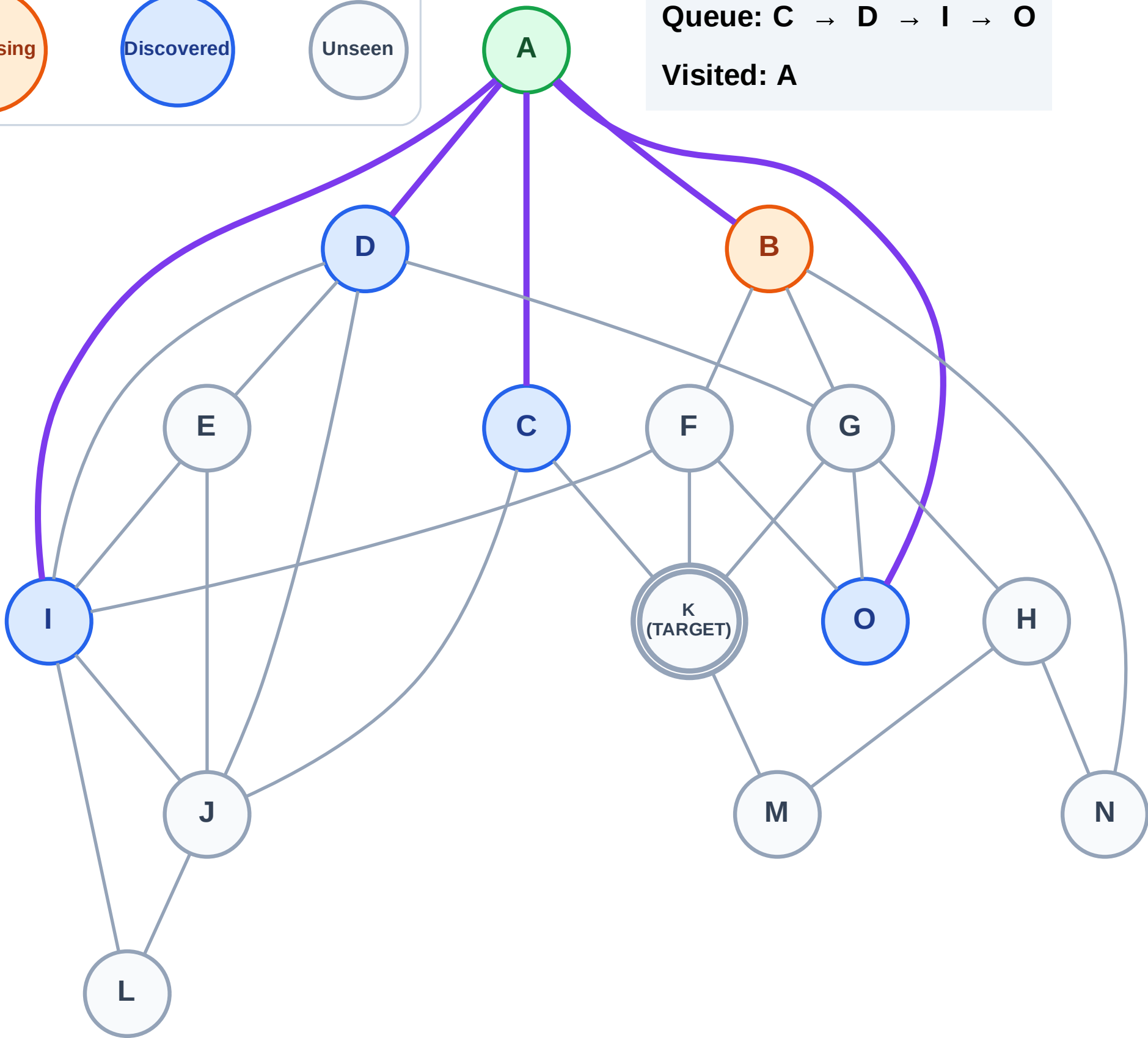
Processing

Discovered

Unseen

Queue: C → D → I → O

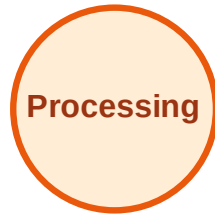
Visited: A



BFS Traversal

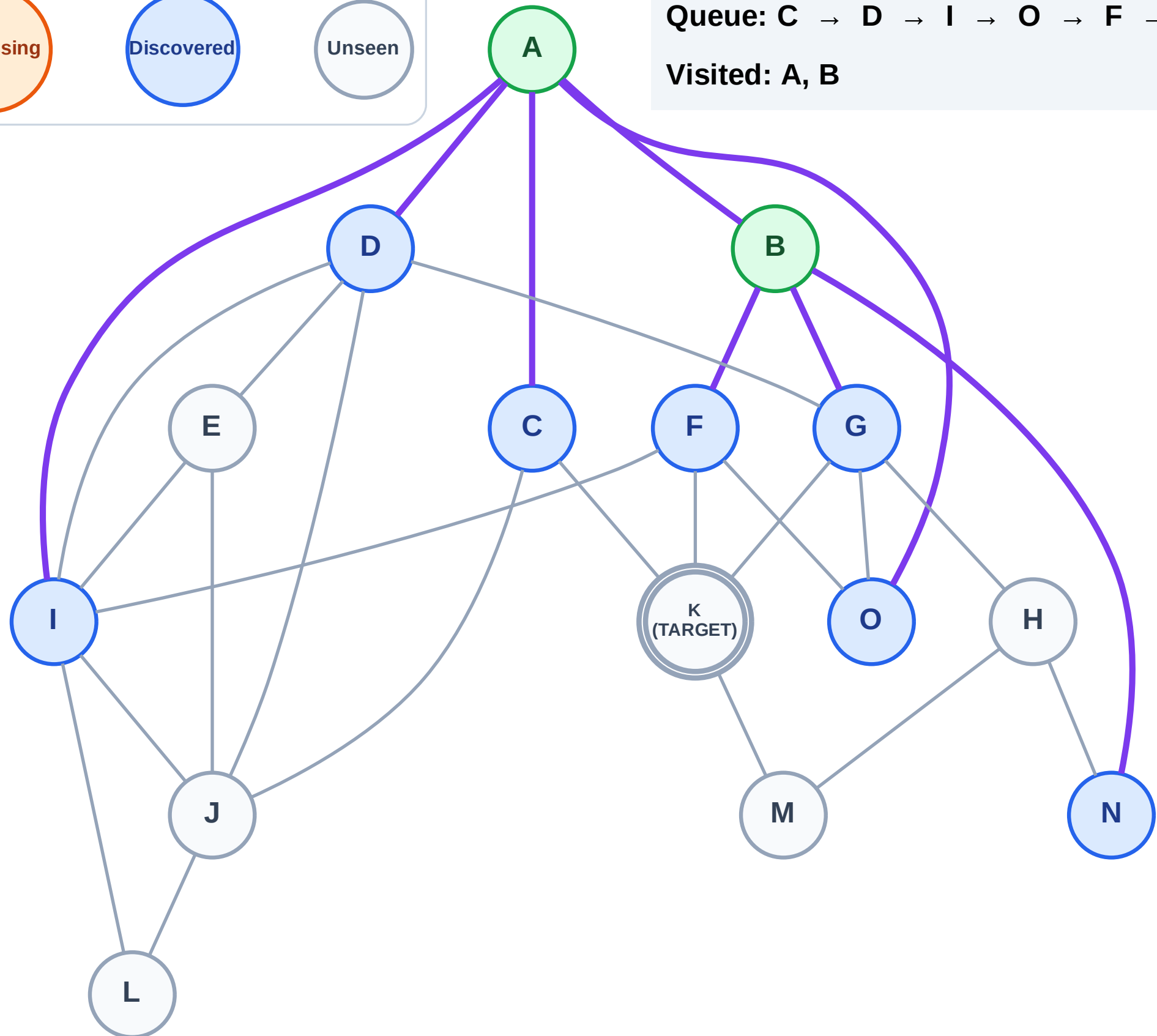
Step 5: Discover F, G, N from B

Legend



Queue: C → D → I → O → F → G → N

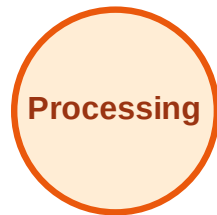
Visited: A, B



BFS Traversal

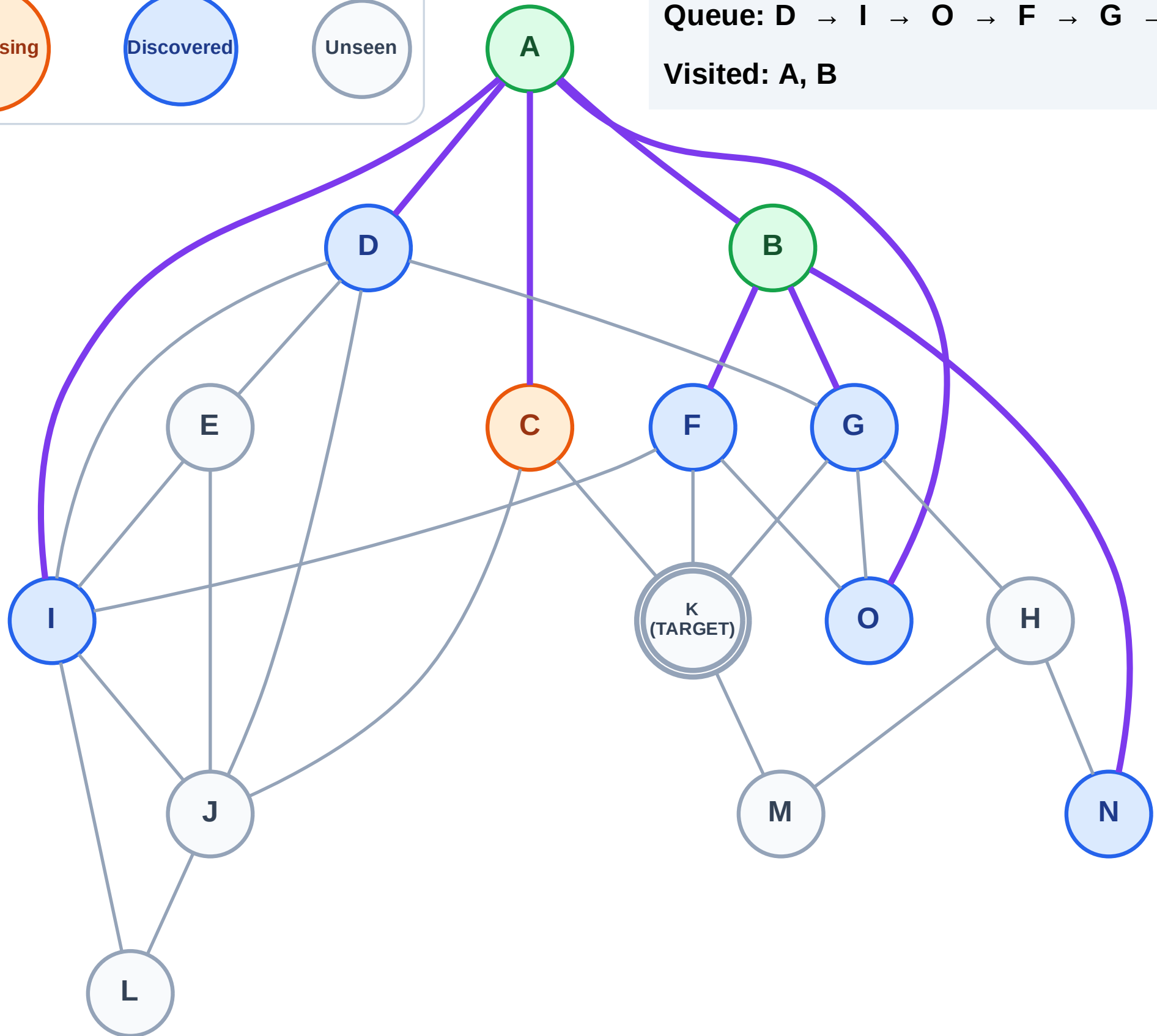
Step 6: Process node C

Legend



Queue: D → I → O → F → G → N

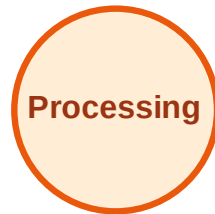
Visited: A, B



BFS Traversal

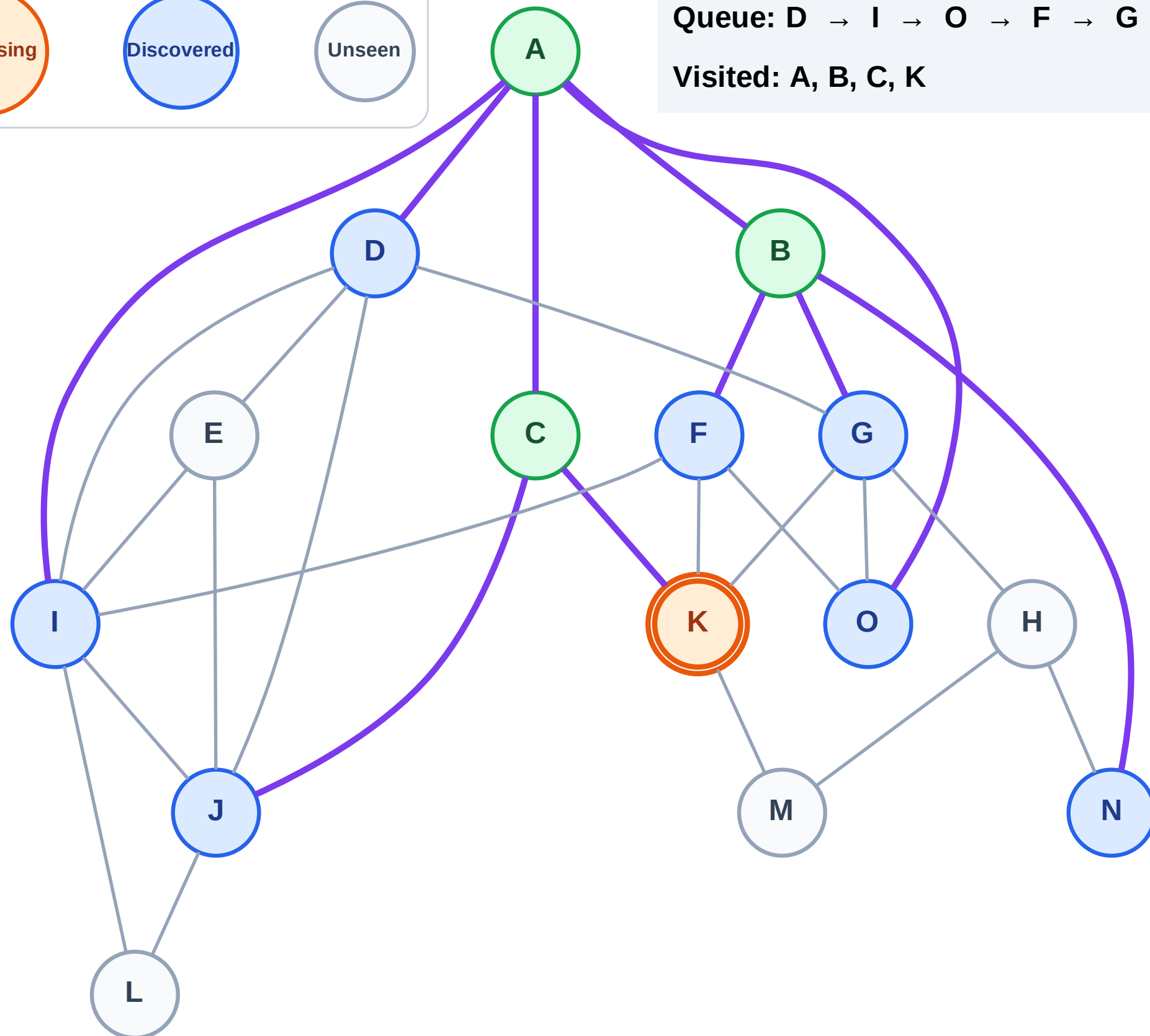
Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → C → K

Legend



Queue: D → I → O → F → G → N → J

Visited: A, B, C, K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

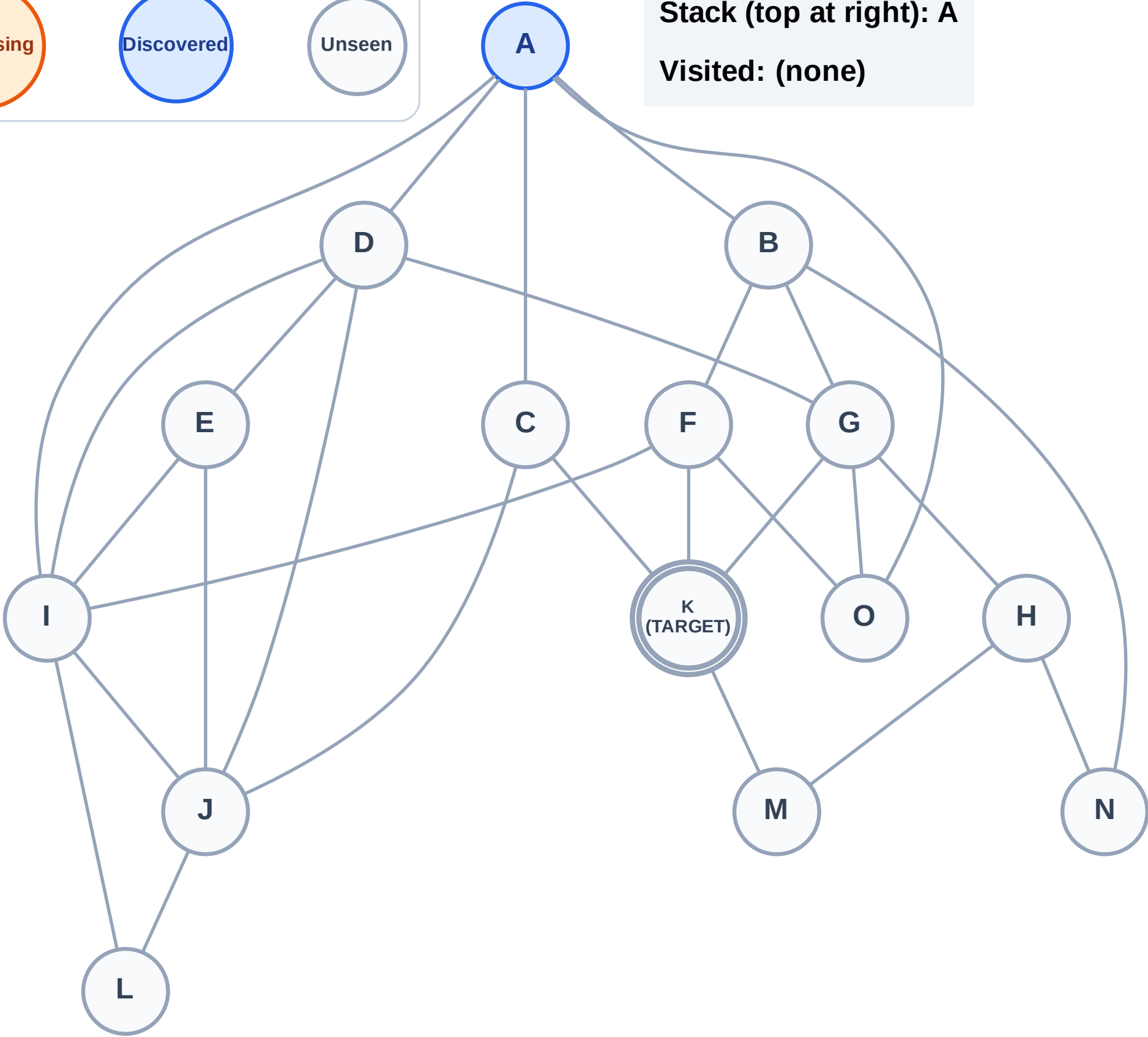
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

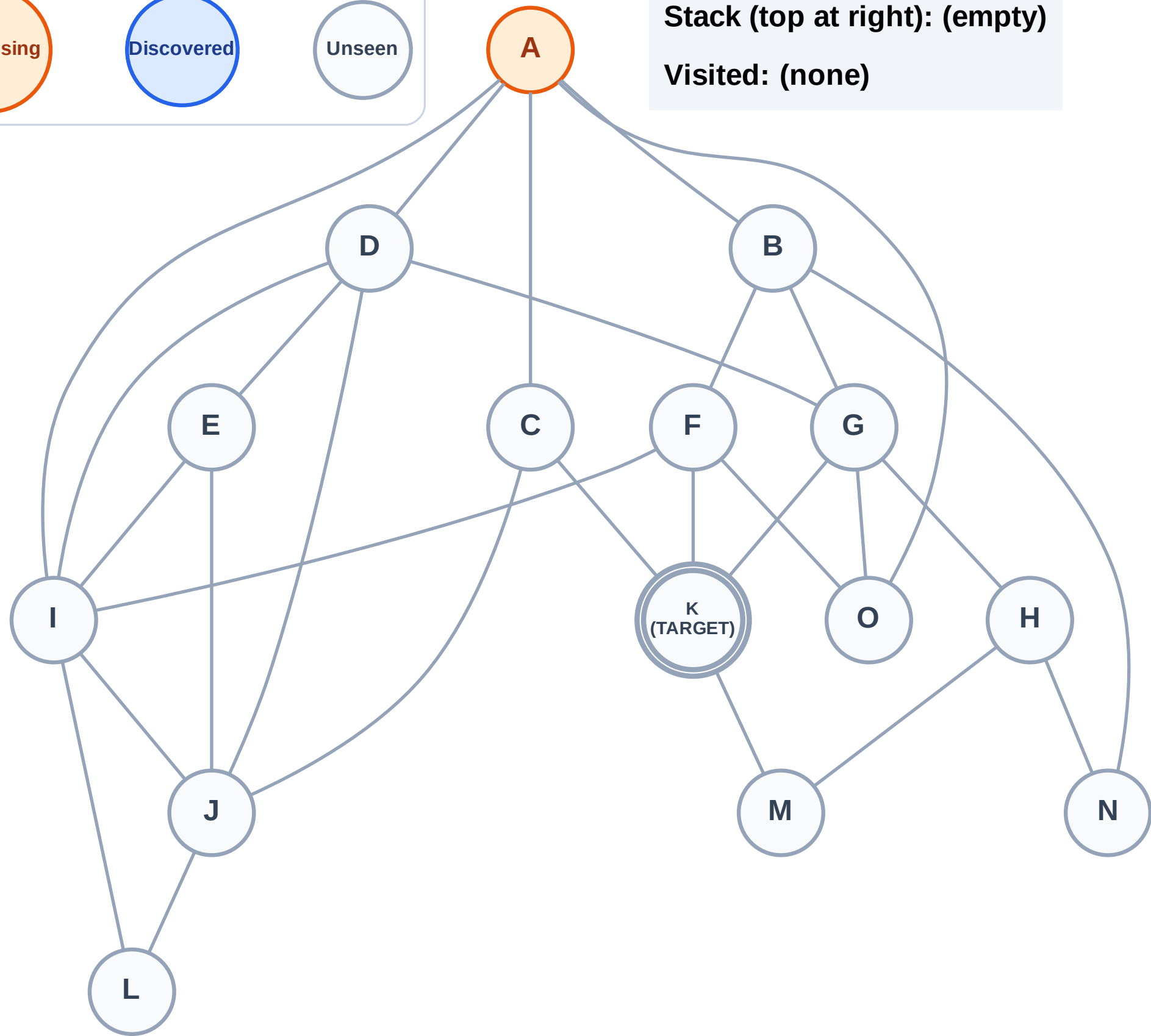
Complete

Processing

Discovered

Unseen

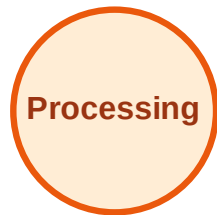
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

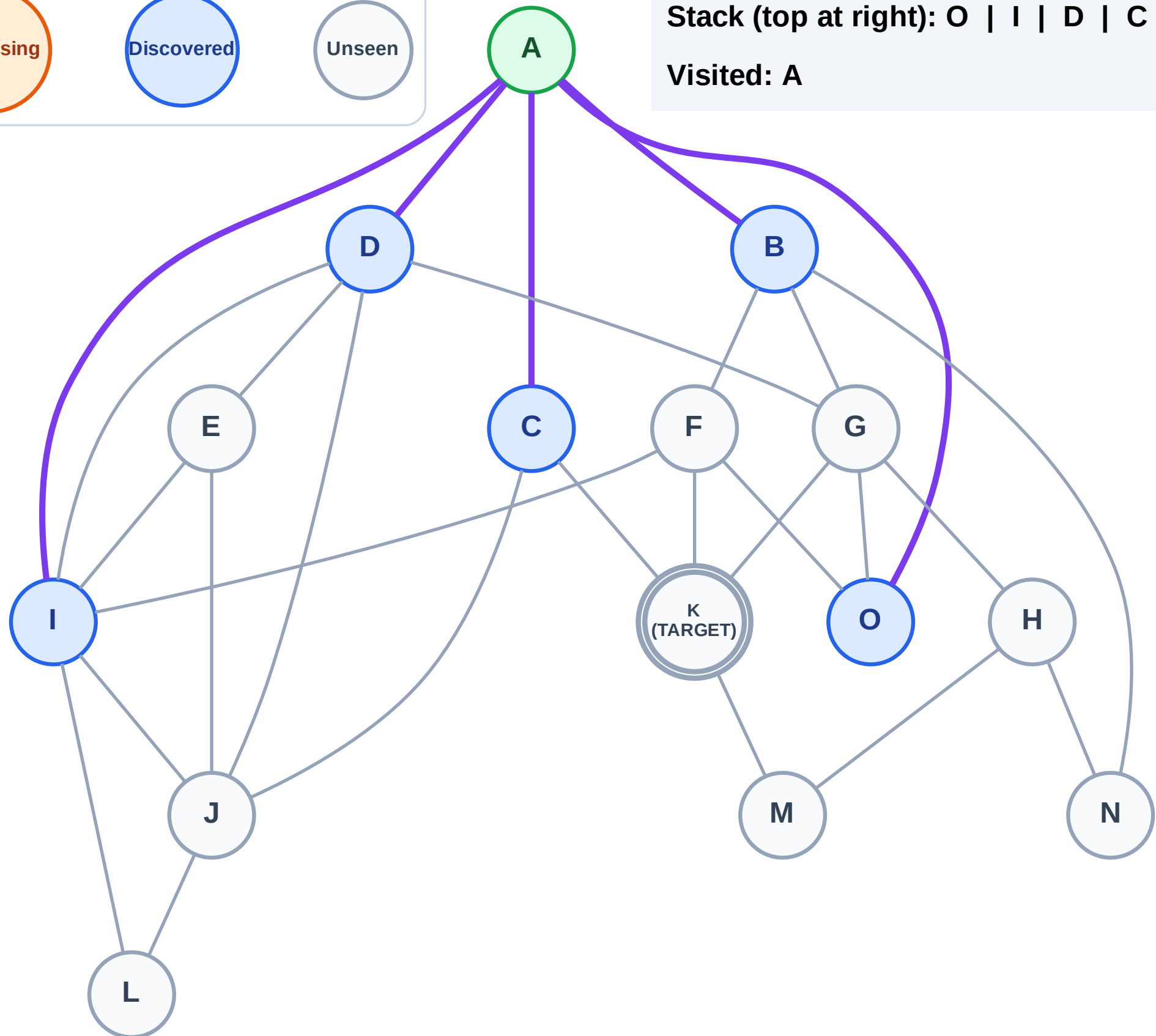
Step 3: Discover O, I, D, C, B from A

Legend



Stack (top at right): O | I | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

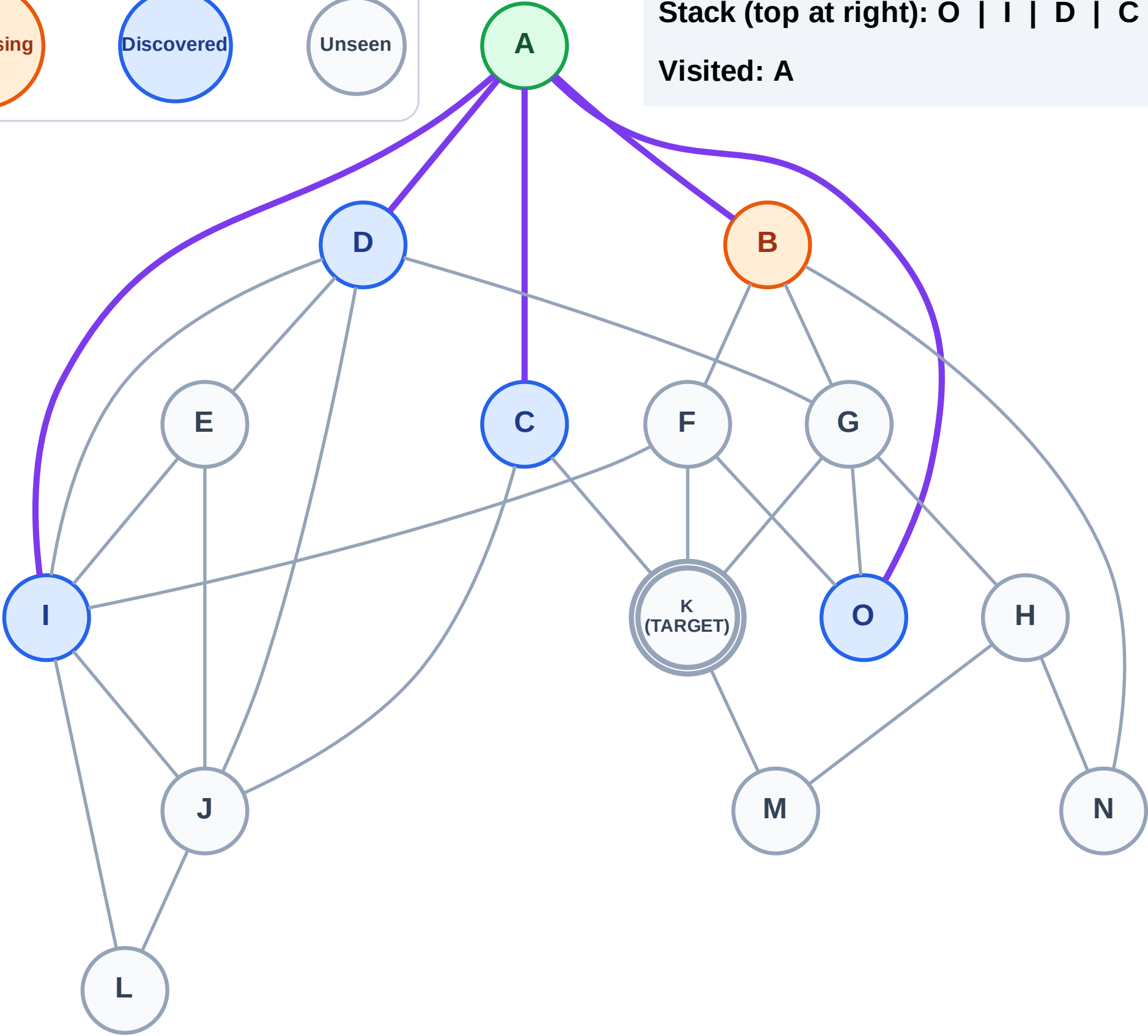
Complete

Processing

Discovered

Unseen

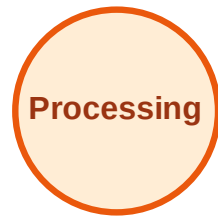
Stack (top at right): O | I | D | C
Visited: A



DFS Traversal

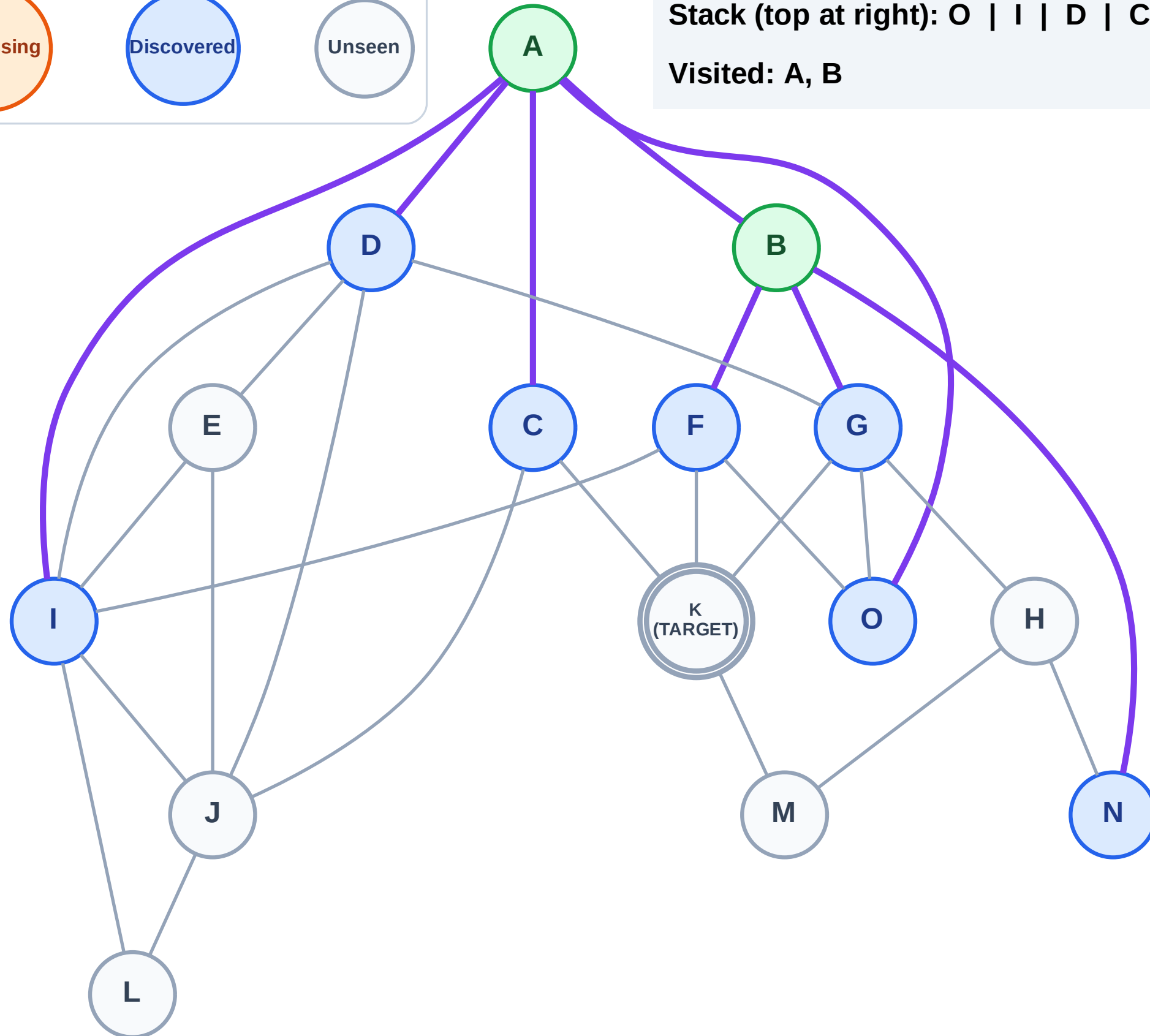
Step 5: Discover N, G, F from B

Legend



Stack (top at right): O | I | D | C | N | G | F

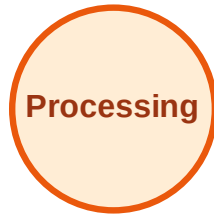
Visited: A, B



DFS Traversal

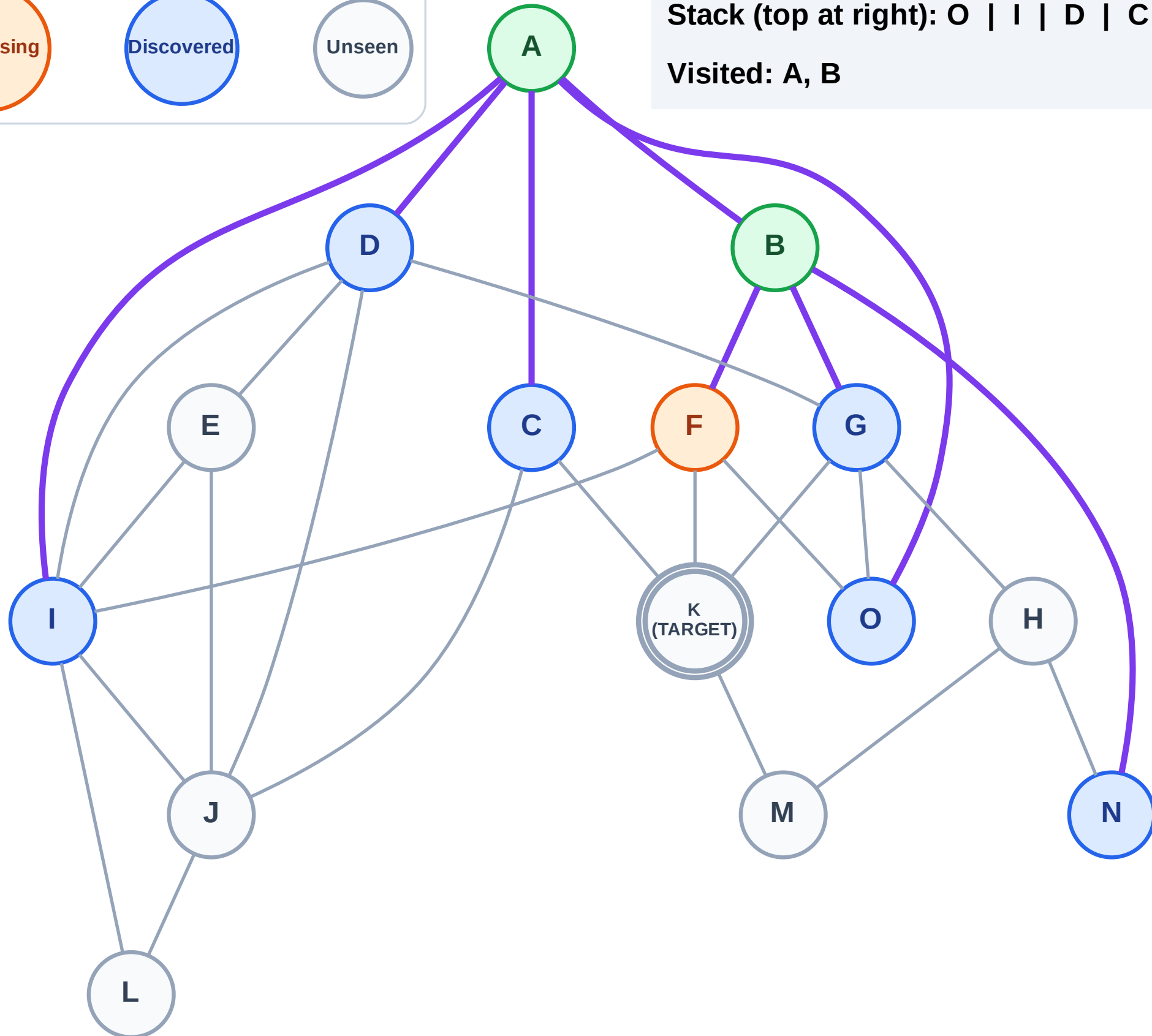
Step 6: Process node F

Legend



Stack (top at right): O | I | D | C | N | G

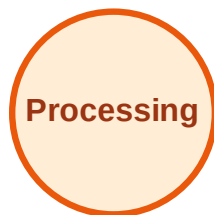
Visited: A, B



DFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → F → K

Legend



Stack (top at right): O | I | D | C | N | G

Visited: A, B, F, K

